Q1 - 2024 (04 Apr Shift 2)

Given that the inverse trigonometric function assumes principal values only. Let x, y be any two real numbers in $[-1,1]$ such that $\cos^{-1} x - \sin^{-1} y = \alpha$, $\frac{-\pi}{2} \leq \alpha \leq \pi$.
Then, the minimum value of $x^2 + y^2 + 2xy \sin \alpha$ is

(1) 0

(2) -1

(3) $\frac{1}{2}$

(4) $-\frac{1}{2}$

Q2 - 2024 (06 Apr Shift 1)

For $n \in \mathrm{N}$, if $\cot^{-1} 3 + \cot^{-1} 4 + \cot^{-1} 5 + \cot^{-1} n = \frac{\pi}{4}$, then n is equal to_____

Q3 - 2024 (09 Apr Shift 2)

The integral $\int_{1/4}^{3/4} \cos\left(2 \cot^{-1} \sqrt{\frac{1-x}{1+x}}\right) dx$ is equal to

(1) $1/2$

(2) $-1/2$

(3) $-1/4$

(4) $1/4$

Q4 - 2024 (09 Apr Shift 2)

Let the inverse trigonometric functions take principal values. The number of real solutions of the equation $2 \sin^{-1} x + 3 \cos^{-1} x = \frac{2\pi}{5}$, is ________

Answer Key

Q1 (1) **Q2** (47) **Q3** (3) **Q4** (0)

Q1

$$\cos^{-1} x-\left(\frac{\pi}{2}-\cos^{-1} y\right)=\alpha$$

$$\cos^{-1} x+\cos^{-1} y=\frac{\pi}{2}+\alpha$$

$$\alpha \in\left[-\frac{\pi}{2}, \pi\right], \frac{\pi}{2}+\alpha \in\left[0, \frac{3 \pi}{2}\right]$$

$$\cos^{-1}\left(x y-\sqrt{1-x^2} \sqrt{1-y^2}\right)=\frac{\pi}{2}+\alpha$$

$$x y-\sqrt{1-x^2} \sqrt{1-y^2}=-\sin \alpha$$

$$(x y+\sin \alpha)=\left(1-x^2\right)\left(1-y^2\right)$$

$$x^2 y^2+2 x y \sin a+\sin^2 a=1-x^2-y^2+x^2 y^2$$

$$x^2+y^2+2 x y \sin \alpha=1-\sin^2 \alpha$$

$$x^2+y^2+2 x y \sin \alpha=\cos^2 \alpha$$

Min. value of $\cos^2 \alpha=0$

At $\alpha=\frac{\pi}{2}$

Q2

$$\cot^{-1} 3+\cot^{-1} 4+\cot^{-1} 5+\cot^{1} n=\frac{\pi}{4}$$

$$\tan^{-1} \frac{1}{3}+\tan^{-1} \frac{1}{4}+\tan^{-1} \frac{1}{5}+\tan^{-1} \frac{1}{\mathrm{n}}=\frac{\pi}{4}$$

$$\tan^{-1}\left(\frac{46}{48}\right)+\tan^{-1} \frac{1}{\mathrm{n}}=\frac{\pi}{4}$$

$$\tan^{-1}\left(\frac{23}{24}\right)+\tan^{-1} \frac{1}{\mathrm{n}}=\frac{\pi}{4}$$

$$\tan^{-1} \frac{1}{\mathrm{n}}=\tan^{-1} 1-\tan^{-1} \frac{23}{24}$$

$$\tan^{-1} \frac{1}{n}=\tan^{-1} 1-\tan^{-1} \frac{23}{24}$$

$$\tan^{-1} \frac{1}{n}=\tan^{-1}\left(\frac{1-\frac{23}{24}}{1+\frac{23}{24}}\right)$$

$$\tan^{-1} \frac{1}{n}=\tan^{-1}\left(\frac{\frac{1}{24}}{\frac{47}{24}}\right)$$

$$\tan^{-1} \frac{1}{n}=\tan^{-1} \frac{1}{47}$$

$$n=47$$

Q3

Solutions **MathonGo**

$$I=\int_{1/4}^{3/4}\cos\left(2\cot^{-1}\left(\sqrt{\frac{1-x}{1+x}}\right)dx\right)$$

$$\int_{1/4}^{3/4}\cos\left(2\left(\tan^{-1}\sqrt{\frac{1+x}{1+x}}\right)\right)dx$$

$$\int_{1/4}^{3/4}\frac{1-\tan^2\left(\tan^{-1}\sqrt{\frac{1+x}{1-x}}\right)}{1+\tan^2\left(\tan^{-1}\sqrt{\frac{1+x}{1-x}}\right)}\Bigg)dx$$

$$=\int_{1/4}^{3/4}\frac{1-\left(\frac{1+x}{1-x}\right)}{1+\left(\frac{1+x}{1-x}\right)}dx=\int_{1/4}^{3/4}\frac{-2x}{2}dx$$

$$=\int_{1/4}^{3/4}(-x)dx=-\left(\frac{x^2}{2}\right)_{1/4}^{3/4}$$

$$=-\frac{1}{2}\left[\frac{9}{16}-\frac{1}{16}\right]$$

$$=-\frac{1}{4}$$

Q4

$$2\sin^{-1}x+3\cos^{-1}x=\frac{2\pi}{5}$$

$$\Rightarrow\quad \pi+\cos^{-1}x=\frac{2\pi}{5}$$

$$\Rightarrow\quad \cos^{-1}x=\frac{-3\pi}{5}$$

Not possible

Ans. 0